

ELEMENT E

STEM concepts support our proposed solution because of the different elements incorporated to create our project. For example, our project is related to engineering and mathematics because it needs to be physically made using specific measurements and criteria to satisfy our stakeholders' demands. This design relates to engineering because our product is being created to solve a problem. This product incorporates mathematics because of the different measurements and dimensions that were used to create it. For example, we considered the formula of the area of a square ($A=a^2$) and the diameter of a circle ($d=2r$) to help create the frame of our first prototype. Mathematics is also incorporated into the interior of our product. This has to do with the spring that operates on the inside of our product. The formula that the spring operates by is $F=-kx$. F is the force in newtons/restoring force, k is the spring constant, and x is the amount of extension. This formula describes how the overall spring operates inside the product. Science is also represented in this product through the structure and function of our design. The structure of our design's interior and exterior tie into the way that it functions.

[Hooke's Law & The Spring Constant | Spring Constant Units & Formula - Lesson | Study.com](#)
[Structure and Function.](#)